

1 The necessity of identity and the necessity of distinctness

NI $\forall xy. x = y \rightarrow \Box(x = y)$

ND $\forall xy. x \neq y \rightarrow \Box(x \neq y)$

- I'm interested in the general versions where the variables x and y can be of any type (so long as the types are the same).
- Take $\Box$ to represent *metaphysical* necessity, or any other non-epistemic interpretation of 'necessarily': I'll be happy if I can defend the view that ND fails on some such interpretation.¹

¹ Epistemic interpretations are also interesting but raise a host of extra complications.

2 The Quine-Kripke argument for NI

A seminal result:²

1. $\forall xy. x = y \rightarrow \Box(x = x) \rightarrow \Box(x = y)$
2. $\forall x. \Box(x = x)$
3. $\forall xy. x = y \rightarrow \Box(x = y)$

² Quine 1953, Kripke 1971. Burgess (2014) investigates the history and concludes that the argument was first formulated by Quine.

Line 1 is the universal generalization of an instance of:

Subst $a = b \rightarrow P[a/v] \rightarrow P[b/v]$ (provided a and b are safely substitutable for v in P)

In a language with lambda predicates can be derived by beta-conversion³ from

LL $a = b \rightarrow Fa \rightarrow Fb$

³ The rule that lets us substitute $(\lambda v.P)a$ with $P[a/v]$ so long as a is safely substitutable for v in P . An "extensional" form of beta-conversion would be sufficient.

3 Williamson's argument for ND

Here is the argument in a streamlined form.⁴ It's given in a language $\mathcal{L}_\Box^\@$ that supplements a basic (higher order) modal language $\mathcal{L}_\Box$ with an additional sentence-forming operator $\@$.

1. $\forall xy. \Box(x = y \rightarrow \@(x \neq y) \rightarrow \@(x \neq x))$ Subst + Normality $_\Box$
2. $\forall xy. x \neq y \rightarrow \Box\@(x \neq y)$ **A1:** $P \rightarrow \Box\@P$
3. $\forall x. \Box\neg\@(x \neq x)$ **A2:** $\Box\neg\@\bot$ + Normality $_\@$
4. $\forall xy. x \neq y \rightarrow \Box(x \neq y)$ Normality $_\Box$, 1–3

⁴ Close to the one in Fritz 2023: 6.2. Fritz's version uses **A2':** $\neg P \rightarrow \Box\neg\@P$ instead of A2. I think this is less clearly motivated by the ordinary practice; but it's equivalent to A2 given A1, Normality $_\Box$, and Normality $_\@$.

Normality $_\@$: $*(P \rightarrow Q) \rightarrow *P \rightarrow *Q$, plus $*P$ whenever P is a theorem (of the logic not including A1 and A2).

4 Doubts about Williamson's appeal to substitutivity

4.1 Intuitive doubts

Setup: a and b are two protons. c and d are two electrons. c and nothing else orbits a . d and nothing else orbits b .

$P := a$ is identical to b and c orbits a and d does not orbit a $P := a = b \wedge Oca \wedge \neg Odb$

- | | |
|---|---|
| 1. If P , every electron that actually orbits a would orbit a . | $P > \forall x.Ex \rightarrow @Oxa \rightarrow Oxa$ |
| 2. If P , every electron that actually orbits b would orbit a . | $P > \forall x.Ex \rightarrow @Oxb \rightarrow Oxa$ |
| 3. If P , d would orbit a . | $P > Odb$ |
| 4. If P , it would not be that P | $P > \neg P$ |

Here, the transition from 1 to 2 seems highly questionable! But this is licensed by application of Subst (under $>$).

4.2 Formal doubts

It's well-known⁵ that simply adding @ is inadequate to capture the range of readings of English sentences that 'actually' can make salient:

⁵ Vlach 1973; Hodes 1984; Cresswell 1990

- (1) If you had had children, then it would have been possible for you not to have had every child you [actually] had.

This has suggested various more flexible formalisms. One⁶ adds two countably infinite families of sentence-modifiers $\{\uparrow_i\}$ and $\{\downarrow_i\}$.⁷

⁶ Cresswell 1990; Correia 2007.

⁷ We treat $\uparrow_i$ and $\downarrow_i$ as syncategorematic formula-builders, not terms of type $t \rightarrow t$ in their own right.

With these, (1) can be formalized as follows:⁸

⁸ C: is a child of; a : you; $\diamond$: it is possible for it to be the case that...; $>$: if it had been that... it would have been that...

- (2) $(\exists x.Cxa) > \uparrow_0 \diamond \neg \forall x. \downarrow_0 Cxa \rightarrow Cxa$

Let $\mathcal{L}^{\uparrow\downarrow}$ be the result of adding these to a given higher-order language $\mathcal{L}$. We can translate sentences of $\mathcal{L}^@$ to sentences of $\mathcal{L}^{\uparrow\downarrow}$ by replacing each @ with $\downarrow_0$ and prefixing $\uparrow_0$.⁹

⁹ An occurrence of $\downarrow_i$ is *free* when it's not in the scope of $\uparrow_i$. A sentence of $\mathcal{L}^{\uparrow\downarrow}$ is one with no free variables and no free $\downarrow_i$.

An analogue of Williamson's argument can be given in $\mathcal{L}^{\uparrow\downarrow}$, using plausible principles for $\uparrow_i$ and $\downarrow_i$:

- | | |
|---|---|
| 1. $\uparrow_0 \forall xy. \Box(x = y \rightarrow \downarrow_0(x \neq y) \rightarrow \downarrow_0(x \neq x))$ | Subst + Normality $_{\Box, \uparrow}$ |
| 2. $\uparrow_0 \forall xy. x \neq y \rightarrow \Box \downarrow_0(x \neq y)$ | V1: $\uparrow_i \forall \vec{v}. P \rightarrow \Box \downarrow_i P$ |
| 3. $\uparrow_0 \forall x. \Box \neg \downarrow_0(x \neq x)$ | V2: $\uparrow_i \forall \vec{v}. \Box \neg \downarrow_i \perp$ + Normality $_{\uparrow}$ |
| 4. $\uparrow_0 \forall xy. x \neq y \rightarrow \Box(x \neq y)$ | 1-3 + Normality $_{\Box, \uparrow}$ |
| 5. $\forall xy. x \neq y \rightarrow \Box(x \neq y)$ | 4 + V3: $P \leftrightarrow \uparrow_i P$, where P has no free $\downarrow_i$ |

But even friends of ND should be suspicious of this system, at least if they endorse

Intensionalism

$$\Box(P \leftrightarrow Q) \rightarrow P = Q$$

Adding Intensionalism will make a complete mess of this system.¹⁰ Choose some intuitively contingently true closed sentence P :

¹⁰ I interpret 'Normality' below as applying to V1–V3, not just the basic logic.

- | | |
|---|--|
| 1. $\vdash \uparrow_0(P \rightarrow \Box \downarrow_0 P)$ | V1 |
| 2. $\vdash \uparrow_0(P \rightarrow \downarrow_0 P = \top)$ | 1, Intensionalism |
| 3. $\vdash \uparrow_0 \Box \uparrow_0 \top$ | Normality $_{\uparrow, \Box}$ |
| 4. $\vdash \uparrow_0(\downarrow_0 P = \top \rightarrow \Box \uparrow_0 \Box \top \rightarrow \Box \uparrow_0 \Box \downarrow_0 P)$ | Subst, Normality $_{\uparrow}$ |
| 5. $\vdash \uparrow_0(P \rightarrow \Box \uparrow_0 \Box \downarrow_0 P)$ | 2-4, Normality $_{\uparrow}$ |
| 6. $\vdash \uparrow_0(P \rightarrow \Box \uparrow_0 P)$ | 5, V1, V2, $\top_{\Box}$, Normality $_{\Box, \uparrow, \downarrow}$ |
| 7. $\vdash P \rightarrow \Box P$ | 6, V3, Normality $_{\Box, \uparrow}$ |

$\top_{\Box} : \Box P \rightarrow P$: this follows from Normality $_{\Box}$ and Intensionalism.

What's gone wrong? A guess: the interaction between $\uparrow_i$ and $\downarrow_i$ is analogous to the interaction between quantifiers and variables they bind, requiring something like the 'safely substitutable' proviso in Subst.¹¹

¹¹ If so, it would be foolish to assume that no parallel problem will not arise within the fragment corresponding to $\mathcal{L}_{\Box@}$!

5 The plural *de re*

A great idea from Bricker (1989): the readings of English modal sentences highlighted by 'actually' can also be expressed using plurals:

Every [actual] F might be G	$\exists xx. (\forall y. y \varepsilon xx \leftrightarrow Fy) \wedge \Diamond \forall y. y \varepsilon xx \rightarrow Gx$	The F things are such that it might be that every one of them is G
Some [actual] F must be G	$\exists xx. (\forall y. y \varepsilon xx \leftrightarrow Fy) \wedge \Box \exists y. y \varepsilon xx \wedge Gx$	The F things are such that it must be that some one of them is G

We can apply this translation-scheme to the sentences from section 4.1:

1. If P , every electron that [actually] orbits a would orbit a $\exists xx. (\forall y. y \varepsilon xx \leftrightarrow Oya) \wedge (P > \forall y. y \varepsilon xx \rightarrow Oya)$
'The electrons that orbit a are such that if P , every one of them would orbit a .'
2. If P , every electron that [actually] orbits b would orbit b $\exists xx. (\forall y. y \varepsilon xx \leftrightarrow Oyb) \wedge (P > \forall y. y \varepsilon xx \rightarrow Oyb)$
'The electrons that orbit b are such that if P , every one of them would orbit b .'

Now there isn't even the appearance of a valid argument from 1 to 2 by (modalized) Subst.

6 Formalizing plural logic within higher-order logic

Elsewhere¹², I proposed that English plural quantification should be interpreted as quantification into predicate type, restricted to the *rigid*.¹³

I'll use $\forall^R, \exists^R$ for higher-order quantifiers restricted to the rigid.

For the translation to respect ordinary reasoning with plurals, we will need the following principle:

Rigid Comprehension $\forall X^\tau. \exists^R U. U \equiv X$

Here $\equiv$ expresses coextensiveness.¹⁴ This can be added to higher-order modal logic¹⁵ without making ND a theorem.

¹² Dorr, Hawthorne, and Yli-Vakkuri 2021: sect. 1.5

¹³ Given an appropriate background higher-order modal logic, this can be defined as follows: **Rigid** $_{\vec{\sigma} \rightarrow t} X := \Box \forall Z. (\forall \vec{y}. X\vec{y} \rightarrow \Box Z\vec{y}) \leftrightarrow \Box \forall \vec{y}. X\vec{y} \rightarrow Z\vec{y}$.

¹⁴ $A \equiv B := \forall \vec{x}. A\vec{x} \leftrightarrow B\vec{x}$.

¹⁵ and even to Classicism: Bacon and Dorr 2024

7 Translations

Can the Bricker-style approach accommodate the full range of meanings $\mathcal{L}^{\uparrow\downarrow}$ aims to regiment? We can answer this question by providing a translation from $\mathcal{L}^{\uparrow\downarrow}$ into $\mathcal{L}$. The details are in the appendix; here, I'll just give some examples to display the main points.¹⁶

Sentences formed by $\downarrow_i$ become predications of a higher order variable bound by a quantifier at the position of the closest $\uparrow_i$ above it, taking

¹⁶ $\Box(\dots)$ in the below examples could be replaced with any arbitrary $\mathcal{L}$ -context.

any variables bound in between the $\uparrow_i$ and the $\downarrow_i$ as arguments:

$$\begin{aligned}
 (3) \quad & \uparrow_0 \quad \square \forall y. \downarrow_0 Oya \rightarrow Oya \\
 & \mapsto \exists^R U. U \equiv (\lambda y. Oya) \wedge \square \forall y. Uy \rightarrow Oya \\
 (4) \quad & \forall x. \uparrow_0 \quad \square \forall y. (\lambda z. \downarrow_0 Oyz)x \rightarrow Oyx \\
 & \mapsto \forall x. \exists^R V. V \equiv (\lambda yz. Oyz) \wedge \square \forall y. (\lambda z. Vyz) x \rightarrow Oyx
 \end{aligned}$$

Variables bound *above* the $\uparrow_i$ are treated like constants:

$$\begin{aligned}
 (5) \quad & \forall x. \uparrow_0 \quad \square \forall y. \downarrow_0 Oyx \rightarrow Oyx \\
 & \mapsto \forall x. \exists^R U. U \equiv (\lambda y. Oyx) \wedge \square \forall y. Uy \rightarrow Oyx
 \end{aligned}$$

The difference between (4) and (5) shows that failures of ND lead to failures of Beta-conversion in $\mathcal{L}^{\uparrow\downarrow}$.

As advertised, instances of Subst do not behave like theorems when embedded:¹⁷

¹⁷ By contrast, instances of LL are still everywhere interchangeable with $\top$.

$$\begin{aligned}
 (6) \quad & \uparrow_0 \quad \square \forall x. x = a \rightarrow \downarrow_0 Fx \rightarrow \downarrow_0 Fa \\
 & \mapsto \exists^R Up. U \equiv (\lambda x. Fx) \wedge p \equiv Fa \wedge \square \forall x. x = a \rightarrow Ux \rightarrow p \\
 & \dashv\vdash Fa \vee \exists^R U. U \equiv F \wedge \square \forall x. x = a \rightarrow Ux \rightarrow \perp \\
 & \dashv\vdash Fa \vee \forall x. Fx \rightarrow \square(a \neq x)
 \end{aligned}$$

Instances of V1–V3 are interchangeable with $\top$, and Normality holds for $\uparrow_i$. But not for $\downarrow_i$: the K axiom $\downarrow_i(P \rightarrow Q) \rightarrow \downarrow_i P \rightarrow \downarrow_i Q$, fails:

$$\begin{aligned}
 (7) \quad & \uparrow_0 \quad \forall x. \square \downarrow_0(Fx \rightarrow \perp) \rightarrow \downarrow_0 Fx \rightarrow \downarrow_0 \perp \\
 & \mapsto \exists^R UVp. U \equiv (\lambda x. Fx \rightarrow \perp) \wedge V \equiv (\lambda x. Fx) \wedge p \equiv \perp \wedge \forall x. \square Ux \rightarrow Vx \rightarrow p \\
 & \dashv\vdash \exists^R UV. U \equiv (\lambda x. \neg Fx) \wedge V \equiv F \wedge \forall x. \square \neg(Ux \wedge Vx) \\
 & \dashv\vdash \forall xy. Fx \wedge \neg Fy \rightarrow \square(x \neq y)
 \end{aligned}$$

When we have the “crossed” binding pattern $\uparrow_i \cdots \uparrow_j \cdots \downarrow_i \cdots \downarrow_j$, the new variable introduced by $\uparrow_i$ will take the one introduced by $\uparrow_j$ as a argument, and will thus be at least third-order:

$$\begin{aligned}
 (8) \quad & \uparrow_0 \quad \square \uparrow_1 \quad \exists x. Fx \wedge \downarrow_0 \diamond \forall y. \downarrow_1 Fy \rightarrow Ayx \\
 & \mapsto \exists^R V. V \equiv (\lambda Ux. \diamond \forall y. Uy \rightarrow Ayx) \wedge \square \exists^R U. U \equiv (\lambda y. Fy) \wedge \exists x. Fx \wedge VUx
 \end{aligned}$$

Our translation-scheme can thus handle more than the similar one defined by Cresswell (1990), which is limited to the second order. It thus escapes Cresswell’s argument against this approach, based on the (alleged) presence of “crossed” readings in English.¹⁸

¹⁸ Cresswell proposes (8) as a regimentation of ‘It ought to be the case that there is a rich person for whom it is actually possible that everyone then rich admires them’. I haven’t managed to find “crossed” examples much more readily intelligible than this.

8 Comparisons

There are two other natural proposals for translating from $\mathcal{L}^{\uparrow\downarrow}$ to $\mathcal{L}$.

- (i) *World-propositions*: replace $\uparrow_i \dots$ and $\downarrow_i \dots$, respectively, with $\exists w_i. (\forall q. q \leftrightarrow \square(w_i \rightarrow q)) \wedge \dots$ and $\square(w_i \rightarrow \dots)$.¹⁹

¹⁹ Where $\{w_i\}$ are fresh variables of type t . This approach is inspired by Fine 1977.

(ii) *Pluralities of propositions*: replace $\uparrow_i \dots$ and $\downarrow_i \dots$, respectively, with $\exists^R X_i. (\forall q. q \leftrightarrow X_i q) \wedge \dots$ and $X_i(\dots)$.²⁰

- On the world-propositions approach, we automatically get V1, V3, Normality $\uparrow$, Normality $\downarrow$, and β -conversion. But adding V2 is equivalent to adding ND.
- On the plurality-of-propositions approach, the situation is the same except that we don't get Normality $\downarrow$ either.²¹
- All three are equivalent given Intensionalism + ND.

²⁰ Where $\{X_i\}$ are fresh variables of type $t \rightarrow t$. This approach is loosely inspired *** by

²¹ I think - I still need to check this.

9 Is this defensible for natural language?

Can we explain the correctness of these translations within a compositional semantic theory for English? What's needed: a constrainable mechanism for putting a $\downarrow_i$ on some predicate, with a corresponding $\uparrow_i$ higher up.²² One approach: use a new *type-raising rule*:

Rigid Lift Any expression that expresses $Y : \sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t$ can also express $\lambda Z^{(\sigma_1 \rightarrow \dots \rightarrow \sigma_n \rightarrow t) \rightarrow t}. \exists^R U. U \equiv Y \wedge ZU$.²³

We can allow this quantification to take scope wherever we like by integrating this with a theory of scope-taking.²⁴

²² Special words like 'actually' are a red herring: they plausibly just highlight readings that would be available without them (Yalcin 2015).

²³ We can control the range of available readings by limiting the application of Rigid Lift. For example, we might limit it to the syntactic category of (possibly modified) *nouns* (e.g. 'electron that orbits *c*'). This would approximately reproduce the predictions of Schwarz's (2012) view that explicit world variables are only associated with determiners.

²⁴ E.g. that of Barker and Shan (2014).

Appendix: the translation function

Following Cresswell (1990), we help ourselves to a collection of new variables Var^+ with countably infinitely variables in each type; let $\mathcal{L}^+$ be the result of adding these new variables to $\mathcal{L}$.²⁵

Since $\mathcal{L}^+$ is still countable, we can define an injective mapping that takes a number n and a term A of $\mathcal{L}^+$ and yields a new variable X_A^n , of the same type as A .

The translations are relativised to a list α of $\mathcal{L}$ -variables and indices.²⁶

- When α is a list, let $\alpha : x$ be the result of appending x to α .
- When α is a list, let $\alpha \setminus x$ be the longest x -free tail of α .
- When α is a list and X is a set, let $\alpha \upharpoonright X$ be the sublist of α that includes only the members of X .
- When B is a term, let $FV(B)$ be the set of all variables that occur free in B .
- When B is a term, let $FV^i(B)$ be the set of all variables of the form X_A^i that occur free in B .

²⁵ Once we know how to translate from $\mathcal{L}^{\uparrow\downarrow}$ into $\mathcal{L}^+$, we can of course translate into $\mathcal{L}$ by replacing members of Var^+ with unused old variables.

²⁶ We assume that the indices on $\downarrow$ and $\uparrow$ are distinct from the variables.

Then our relativized translation function is defined as follows:

$$\begin{aligned}
 v^\alpha &:= v && v \text{ a variable} \\
 c^\alpha &:= c && c \text{ a constant} \\
 (AB)^\alpha &:= A^\alpha B^\alpha \\
 (\lambda v. A)^\alpha &:= (\lambda v. A^{\alpha \cdot v}) \\
 (\downarrow_i P)^\alpha &:= X_{(\lambda v_1 \dots v_n. P^\alpha)}^i v_1 \dots v_n && \text{where } (\alpha \setminus i) \upharpoonright FV(P^\alpha) = [v_1, \dots, v_n] \\
 (\uparrow_i P)^\alpha &:= \exists^R X_{A_1}^i \dots X_{A_n}^i. X_{A_1}^i \equiv A_1 \wedge \dots \wedge X_{A_n}^i \equiv A_n \wedge P^{\alpha \cdot i} && \text{where } FV^i(P^\alpha) = \{X_{A_1}^i, \dots, X_{A_n}^i\}
 \end{aligned}$$

Finally, we define the translation of a sentence P to be $P^\square$.

10 NOTES

What's valid in this system? $\uparrow_i(\neg P \rightarrow \Box \neg \downarrow_i P)$: good. No variables are bound between the $\downarrow_i$ and $\uparrow_i$, so the translation is

$$\exists^R q. q \equiv \neg P \wedge \neg P \rightarrow \Box \neg q$$

which is a logical truth.

$\uparrow_i((\forall \vec{v}. \neg P) \rightarrow (\forall \vec{v}. \Box \neg \downarrow_i P))$: good. Assume for simplicity that all of $\vec{v}$ have a free occurrence in P . Then the translation is

$$\exists^R U. U \equiv (\lambda \vec{v}. P) \wedge ((\forall \vec{v}. \neg P) \rightarrow (\forall \vec{v}. \Box \neg U \vec{v}))$$

this is a theorem: if U is rigid and coextensive with an empty relation, then U is necessarily empty (by Inextensibility).

$\uparrow_i(\forall v. \neg P \rightarrow \Box \neg \downarrow_i P)$: bad. The translation is

$$\exists^R U. U \equiv (\lambda v. P) \wedge \forall v. \neg P \rightarrow \Box \neg Uv$$

Note that by contrast if we put the $\forall v$ outside the $\uparrow_i$, it's good (since it's the universal generalization of something we already showed to be a tautology). So this example doubles as showing that $\uparrow$ doesn't obey BF.

$\uparrow$ does obey normality (for the base logic). If P is a theorem of the base logic, then P^α is a theorem for every list α .

Hmm - given the way I've done it, getting the K axiom for $\uparrow$ turns on Rigid Uniqueness! But if I use universal instead of existential quantifiers, it won't require that. That seems a good reason to redo it that way. Establishing D : $\uparrow_i P \rightarrow \neg \uparrow_i \neg P$ will also be easy on this approach. (Note that this is equivalent to $\neg \uparrow_i \perp$ given Normality for $\uparrow$). But the converse implication (D_c) still requires Rigid Uniqueness.

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